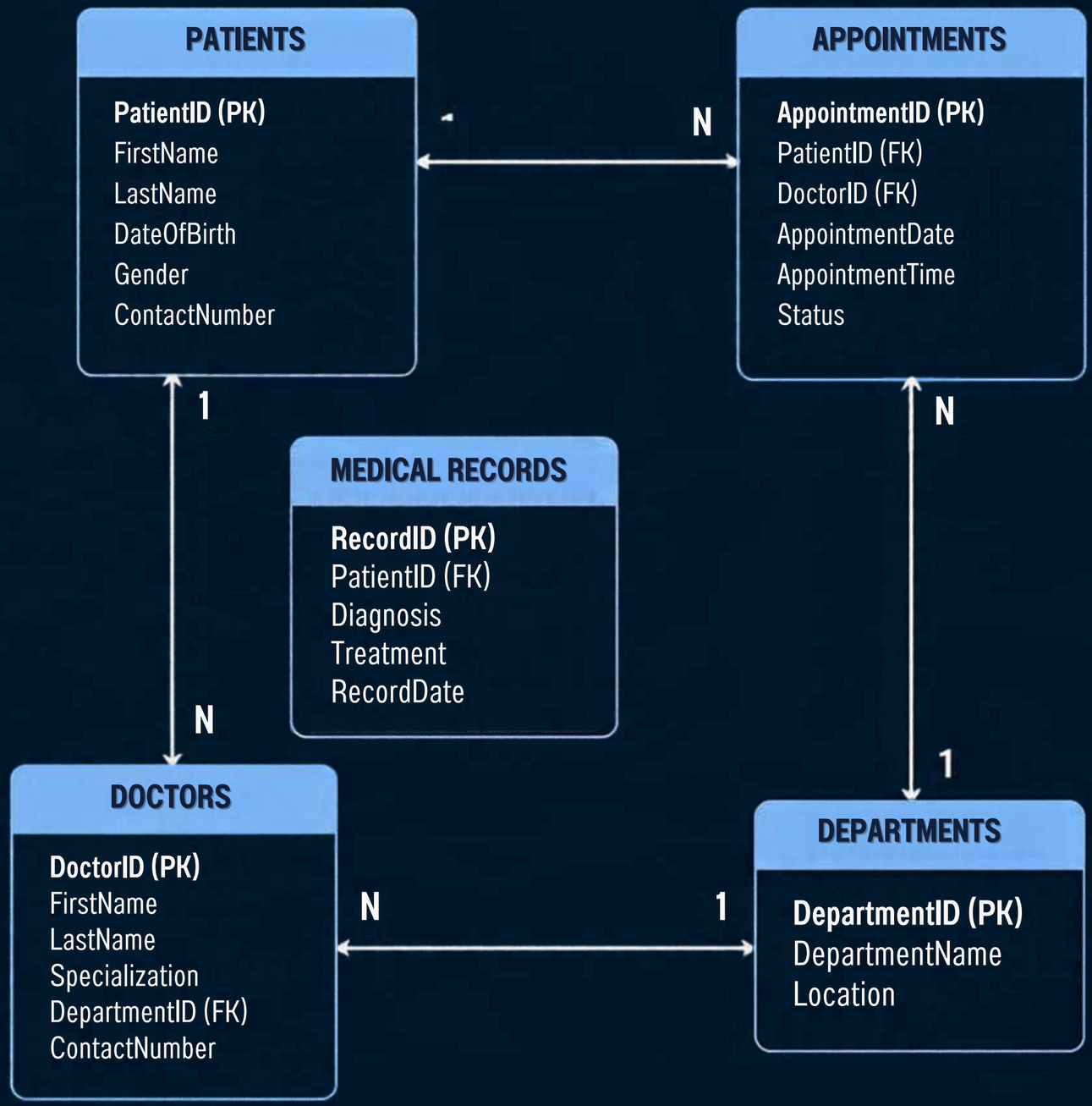
DATABASE SCHEMA OVERVIEW

The core entities that power **appointment** and **patient** care management

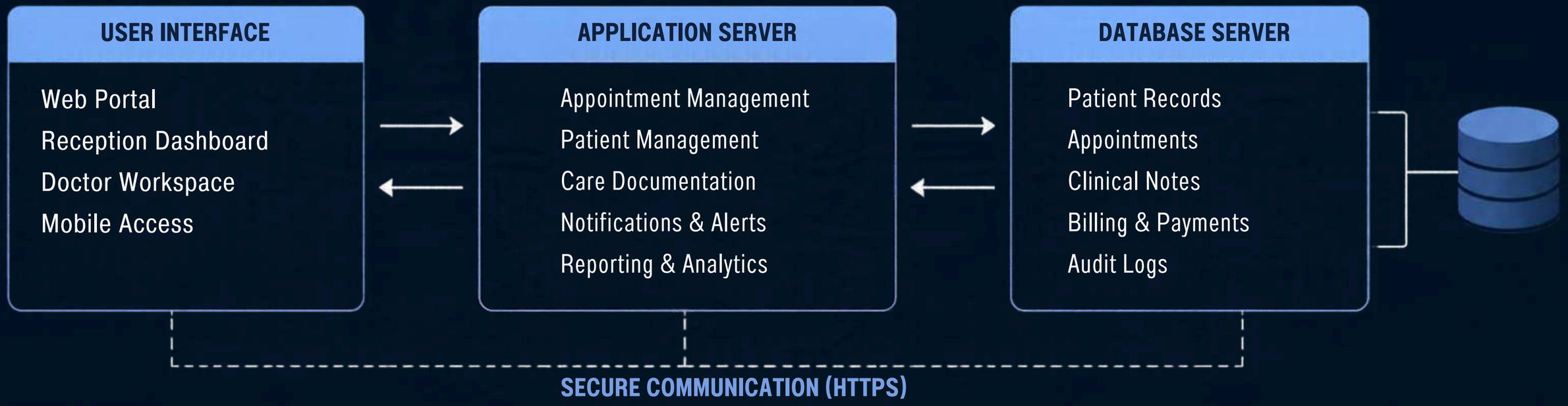
Structured relationships ensure data **integrity** and **consistency**

Supports efficient **scheduling**, **record keeping**, and **reporting**

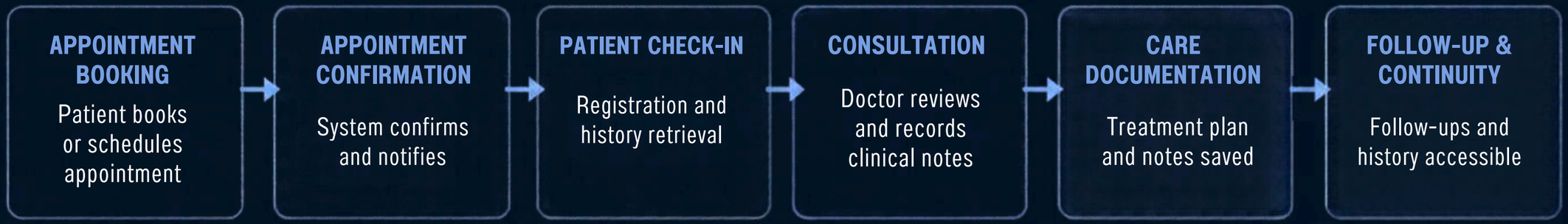


SYSTEM ARCHITECTURE

A secure, **multi-tier architecture** that connects users, applications, and data to power seamless patient care.

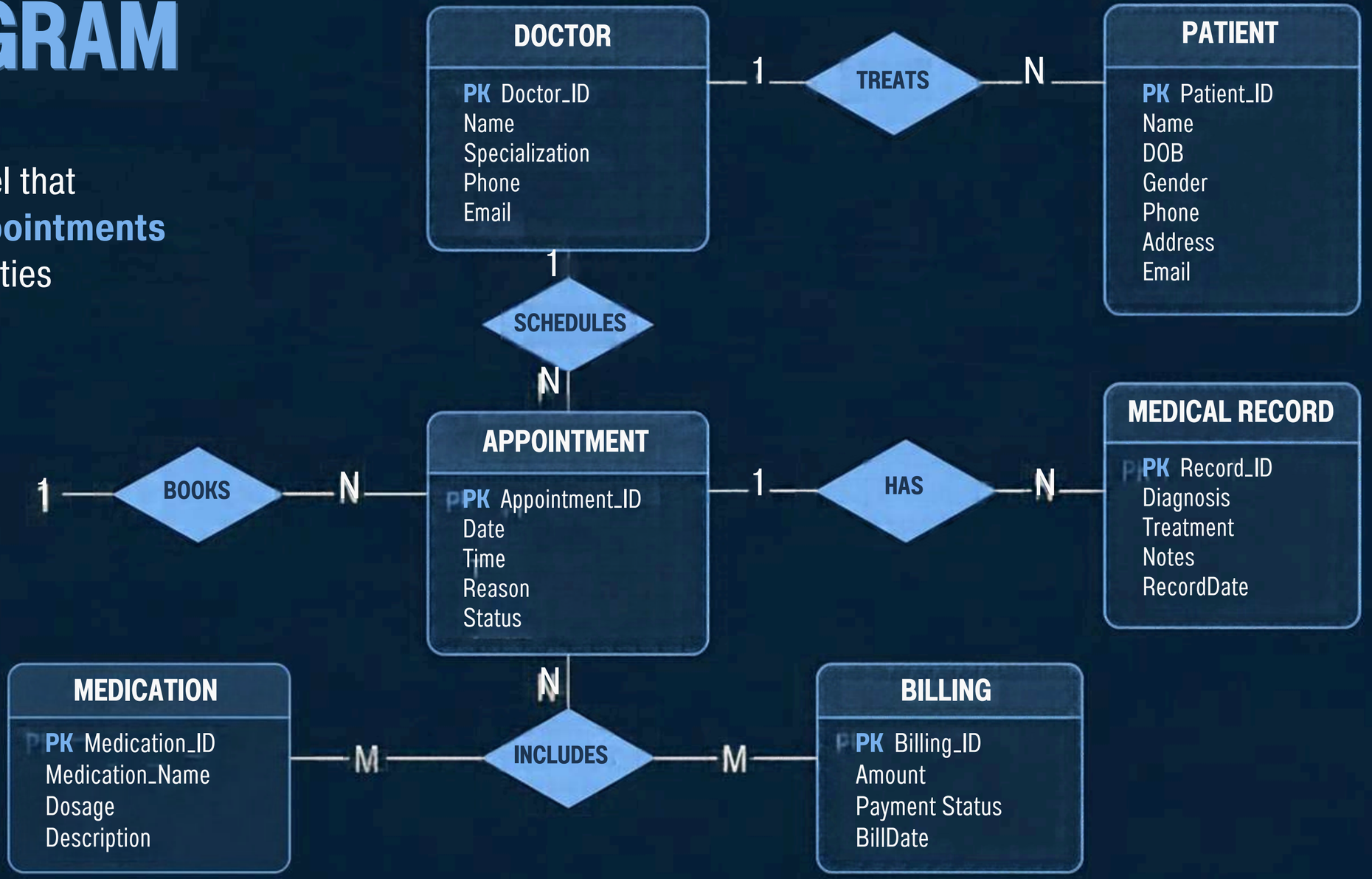


END-TO-END WORKFLOW



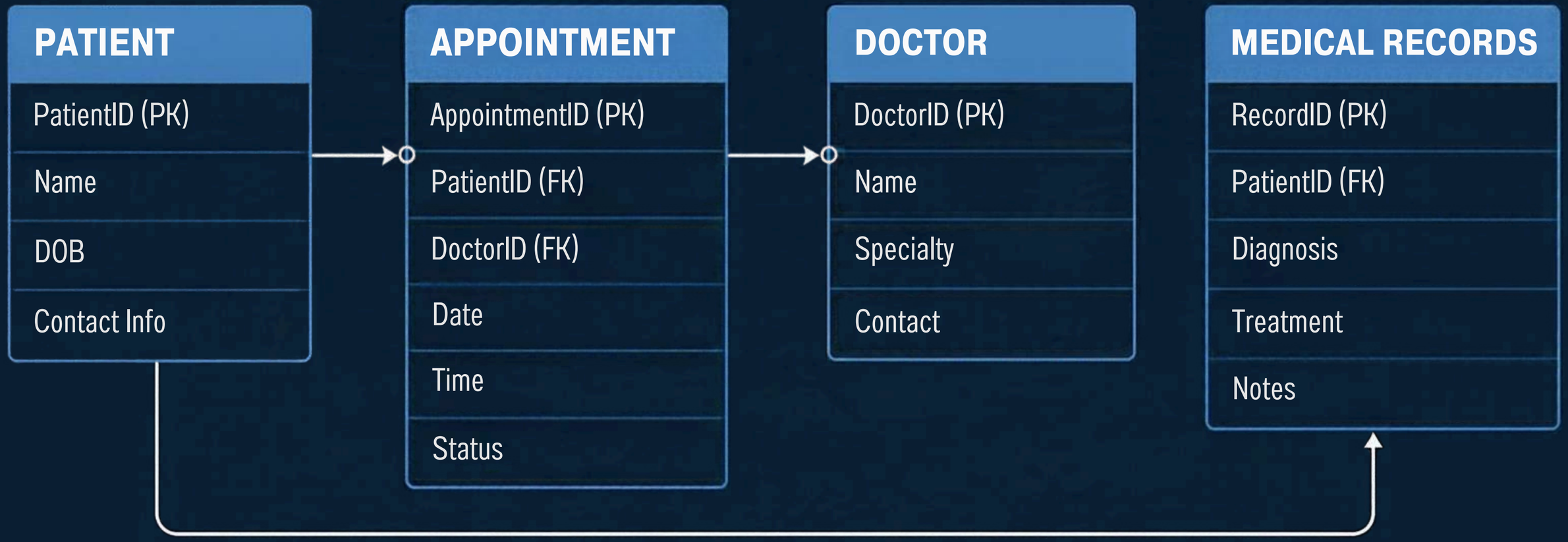
ENTITY-RELATIONSHIP (ER) DIAGRAM

A relational data model that connects hospital **appointments** and **patient care** entities



RELATIONAL TABLES

Core tables and key fields of the hospital management database



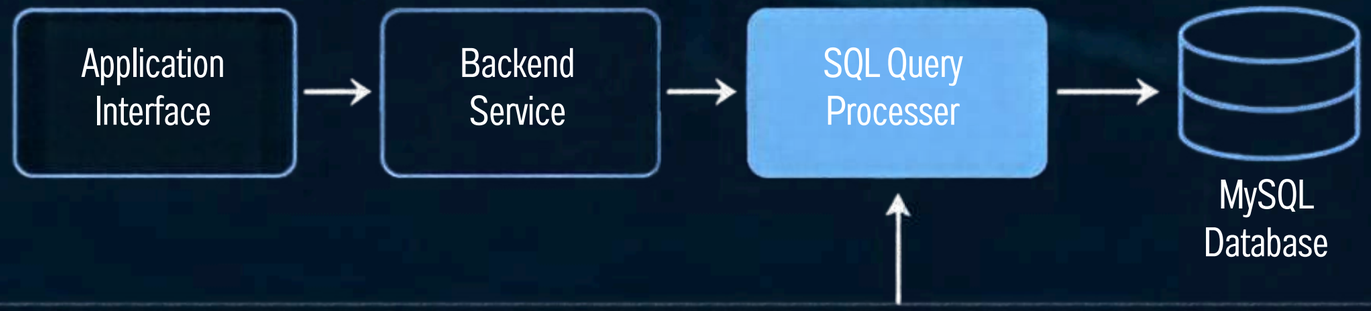
IMPLEMENTATION DETAILS

Built on **MySQL** database platform

Backend developed using **Java/Python** with modern frameworks

Optimized SQL for fast, reliable data access

Scalable architecture for secure, high-performance operations



Appointment Booking Query

```
INSERT INTO appointments
(patient_id, doctor_id, appointment_date,
appointment_time, reason, status)
VALUES (?, ?, ?, ?, ?, ?);

SELECT LASTInsert_ID( )
    AS appointment_id;
```

Record Retrieval Query

```
SELECT a appointment_id, a. appointment_date,
a.appointment_time, d.name AS doctor_name, a/status,
a reason
FROM appointments a
JOIN doctors d ON a.doctor_id = d.doctor_id
WHERE a patient_id = ?
ORDER BY a.appointment_date DESC;
```

Efficient. Secure. Scalable.

Powering seamless appointment and patient care management.

SECURITY AND TESTING

Secure access with user **authentication** and **role-based access control**

Data **encryption** to protect sensitive patient information

Rigorous testing with **unit tests** and **integration tests**

User **acceptance testing** to ensure reliability and real-world readiness



FUTURE ENHANCEMENTS AND CONCLUSION

Expand with **mobile app access** for patients

Leverage **AI** for smarter appointment scheduling

Strengthen **analytics** to drive better patient care

A scalable system that **improves efficiency**, **elevates care**, and **empowers learning**,

